

AI Engineer — Take-Home Test Task

The Scenario

You're building the decision core of a budget allocation system for a performance marketing team. The team manages paid media spend across 12 campaigns on 3 platforms (Google, Meta, TikTok). Every week, a strategist reviews performance and decides how to redistribute a fixed weekly budget across campaigns.

The problem: the strategist's decision logic lives entirely in their head. It's a mix of financial intuition, platform-specific rules, and business constraints that have never been written down formally. Your job is to build a system that can replicate the quality of that reasoning — and be honest when it can't.

What the Candidate Receives

- A JSON file with 12 campaigns, each containing: platform, current weekly spend, ROAS, 4-week ROAS trend, CPA, conversion volume, campaign type (prospecting / retargeting / brand), audience saturation signal (0–1), and platform-level budget cap
 - A natural language "strategy brief" (1 page, written like a real client would write it — imprecise, occasionally contradictory, with priorities that conflict under certain conditions)
 - A total weekly budget: **\$85,000**
 - 3 labelled allocation decisions from previous weeks (ground truth for the eval harness)
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The Task

Build a system that ingests the campaign data and the strategy brief, reasons through a budget reallocation, and outputs a structured allocation plan with justification.

The system must honour three financial principles throughout. These principles define what a correct output looks like — how you implement them is your call.

Principle 1 — Opportunity Cost

Every allocation decision carries a cost: the return you give up by not putting that dollar somewhere else. A correct output reflects this. Headline ROAS alone is not sufficient basis for an allocation decision — the marginal return on the next dollar matters more than the average return to date.

Concretely: a campaign with strong ROAS but a high audience saturation signal is not the same investment as a campaign with the same ROAS and room to scale. The output must distinguish between the two and explain what is being given up in each allocation decision.

What the output must show: for each campaign, a clear statement of the trade-off being made — not just the recommended number.

Principle 2 — Diversification vs. Concentration

The strategy brief contains an instruction along the lines of: *"Focus spend on what's working."* This is intentionally ambiguous. A correct output does not take this instruction literally without examining what concentration risk it creates.

The output must include a **concentration score** for the proposed allocation — something analogous to a Herfindahl–Hirschman Index: the sum of squared spend shares across campaigns. A perfectly concentrated portfolio scores 1.0; a perfectly even distribution scores $1/n$. If the proposed allocation exceeds a concentration threshold (suggested: $HHI > 0.25$), the output must flag the platform and campaign-type risk this creates, and present an alternative diversified allocation alongside the primary recommendation so a human can compare both with trade-offs stated explicitly.

The system should not refuse to concentrate when the data supports it — it should concentrate *and* tell the human what they're betting on.

What the output must show: a concentration score, a warning if the threshold is breached, and a second allocation option when it is.

Principle 3 — Risk Management

Three risk signals must be factored into every allocation decision, not treated as footnotes.

Volatility: Each campaign carries a 4-week ROAS trend. A declining trend — even on a currently strong campaign — changes the risk profile of that position. The output must not treat a campaign with ROAS 3.8 and a declining trend the same as a campaign with ROAS 3.8 and a flat or rising trend.

Platform concentration: No single platform should receive more than 55% of total budget unless the output explicitly justifies the exception. Breaching this threshold is a risk, not a default.

Minimum viable spend: Some campaigns will be below their effective minimum spend threshold (provided in the data). Allocating to a campaign at a level that makes it statistically unviable is a risk. The output must either fund the campaign properly or recommend pausing it and reallocating the freed budget. Half-funding is not a neutral choice.

What the output must show: a risk flag per campaign for each of these three signals where applicable, and a rationale for how the allocation handles or accepts each flagged risk.

The Eval Harness (Required)

Build your own evaluation framework. Three previous allocation decisions are provided as ground truth. The harness must score the system's output on:

- **Allocation accuracy:** within $\pm 10\%$ of ground-truth spend per campaign
- **Action agreement:** does the system agree on scale/hold/reduce/pause?
- **Risk flag recall:** did the system catch the same risk flags a human identified?
- **Total budget constraint:** hard fail if total $\neq$ \$85,000

Define the overall scoring rubric and document it. Minimum bar: pass at least 2 of 3 ground-truth cases across all four dimensions.

Stretch Goals

Constraint correction: When the output allocation doesn't sum to \$85,000, the system must automatically correct it using a deterministic fallback — not simply retry.

Conflict resolution: For each entry in `unresolved_conflicts`, generate a clarifying question ranked by how much the answer would change the allocation. This simulates a human-in-the-loop intake flow.

Deliverables

- Working code in a GitHub repo or ZIP

- README covering: setup, design decisions, which financial assumptions are made explicitly vs. inferred, and what you would do differently with more time
 - Eval harness output showing scores against the 3 ground-truth cases
 - A 15-minute walkthrough call after submission
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What We're Really Testing

The financial principles are a proxy for the actual skill we need: **can this person build a system that reasons under ambiguity, is honest about uncertainty, and surfaces trade-offs rather than hiding them?**

There is no prescribed architecture. The interesting part of the walkthrough is hearing *why* you made the choices you did — where you used deterministic logic, where you reached for a model, and whether those choices reflect genuine judgment about the nature of each sub-problem.

A system that produces confident numbers without modelling opportunity cost or flagging concentration risk has failed — even if the numbers look reasonable. A system that says "I'm concentrating spend on Google and here's what you're betting on" has passed.

A Note on Data Design

When you build the mock campaign data, make it adversarial enough to force real decisions:

- At least 2 campaigns with strong ROAS but high saturation signals
- At least 1 campaign below minimum viable spend
- The top 3 performers on the same platform, so platform concentration trips the 55% threshold
- A declining 4-week trend on at least 2 high-ROAS campaigns
- One instruction in the strategy brief that directly contradicts another

If the data doesn't force the principles to engage, you won't learn anything from the submission.